

Payroll v4 — client-owned rules, zone efficiency, and a spreadsheet grid

Audience: payroll / operations stakeholders · Admin live on production · **No rider app update required** · Module: Payroll & Requests

1. What you requested (MGGO Payroll SOP v4.0)

Payroll in v3 classified a day with one hard-coded rule. The SOP makes the **client** the unit of the calculation, so each client can keep its own arrangement without a code change.

- **The client owns the rules.** Each client keeps its own hour values, thresholds, a default day status, and its own ordered list of day rules.
- **Rules decide a day.** A rule matches on **zone band** (Good / Average / Low), **zone name**, **orders** and **worked hours**; the **first match from the top wins**.
- **The SOP's own result codes:** Full day 12 h, Reduced day 3 h, Half day 6 h, Actual hours, Absent, Absent LH, Absent LO, and Custom hours.
- **Zone efficiency comes from the previous completed month only** — $DPD = \text{orders} \div \text{rider-days}$, $\text{efficiency} = DPD \div \text{the target DPD}$, banded Good / Average / Low.
- **A stated order of precedence:** hand adjustment → Operations (OFF / Sick / Accident / Vehicle issue) → first matching client rule → client default → the old classification.
- **Off Structure per rider** — contracted OFF days for the month (default 2) — drives Required Hours, and must be uploadable in bulk.
- **Excel-style grid:** select a block of rider-days, fill, copy and paste across the block, and adjust the whole selection with **one reason**.
- Attendance Efficiency = $\text{Actual Worked Hours} \div \text{Required Hours}$, uncapped, with status filters.
- Everything above editable by your own staff — no developer, no deployment.

2. What we delivered

- **Payroll is now data, not code.** Clients, their hour values, their default status and their rule lists are stored in the database and edited on screen.
- **Two SOP clients are seeded and ready:** **Americana** (zone band + orders) and **Keeta** (orders + hours), each with the SOP starting rule list for the current month.
- **A stated precedence in one place** — a hand adjustment always beats a rule, and Operations always beats a rule, in the grid, the export, the Attendance & Orders tab and the simulator alike.
- **Zone efficiency from last month**, with a **Recompute** button and an Ops override for DPD, target and band that never destroys the automatic figure.
- **Off Structure** per rider, plus bulk upload from an Excel sheet with a downloadable template and a preview that says what will apply and what will be skipped.
- **The Combined grid behaves like a spreadsheet** — drag to select, corner handle to fill, copy / paste / fill down, and a *reason is required* before any hand adjustment lands.
- **A manual changes record** — every hand adjustment keeps the original value, the new value, the reason, the user and the time, and a later recalculation never overwrites it.
- **A change log** for every rule, threshold, band override and client edit, with your name and the time.
- **Four pages:** Payroll · Combined Payroll · Attendance & Orders · Settings (plus Requests).

3. How a day is decided, in order

#	Source	What it means
1	Hand adjustment	A named person changed that rider-day and left a reason. Nothing overrides this.
2	Operations	An approved OFF / Sick / Accident / Vehicle-issue request covers the day. The day is that status and credits 0 hours.
3	First matching client rule	The client's own list, read top to bottom. The first rule whose conditions all hold decides the day.
4	Client default	No rule matched. The client's own default status applies (Americana: Full day 12 h · Keeta: Actual hours).
5	Previous behaviour	Only when the client has <i>no saved rules at all</i> for that month: a check-in is a full 12 h day, a covered day is that cover, anything else is Absent.

Result codes and the hours they credit: **Full day · 12 h** → 12 h · **Reduced day · 3 h** → 3 h · **Half day · 6 h** → 6 h · **Actual hours** → the logged check-in → check-out hours · **Custom hours** → the number typed · **Absent / Absent LH / Absent LO** → 0 h. Hours are the client's own values, so they can differ per client.

4. Zone efficiency — what the band means

DPD = zone orders ÷ rider-days · **Target DPD** = the average DPD across the zones that actually had rider-days · **Zone efficiency %** = DPD ÷ Target DPD × 100.

It is always read from the **previous completed month**. That is deliberate: a band that moved with the current, half-finished month would silently re-price days you have already decided.

Zone efficiency	Band	Default threshold
at or above	Good	110%
at or above	Average	70%
below	Low	< 70%
no previous-month data	Not set	—

A zone nobody measured is **Not set**, never "Low". A zone with no rider-days has no DPD to average, so it is left out of the target instead of dragging it down. A zone that is Not set matches no seeded rule, so its days fall to the client default — an unmeasured zone is never punished for being unmeasured.

Worked example (Kuwait zones)

Zone	Orders	Rider-days	DPD	Efficiency	Band
Zone A	1,200	40	30.00	130.4%	Good
Zone B	400	25	16.00	69.6%	Low
Zone C	0	0	—	—	Not set

Target DPD = (30.00 + 16.00) ÷ 2 = **23.00**. Had the empty Zone C been counted as 0, the target would have fallen to 15.33 and Zone B would have read Average — which is exactly the distortion the rule avoids.

5. The SOP starting rules, as seeded

Read top to bottom. The first line whose conditions all hold is the answer.

Americana — zone band + orders

Keeta — hours + orders

#	If	Then	#	If	Then
1	Orders < 1	Absent	1	Hours < 0.5 <i>and</i> orders < 1	Absent
2	Zone is Khiran	Full day 12 h	2	Orders < 3	Absent LO
3	Band Low <i>and</i> orders < 5	Reduced day 3 h	3	Hours < 4	Absent LH
4	Band Good or Average <i>and</i> orders < 7	Reduced day 3 h	4	Hours < 6 <i>and</i> orders ≥ 6	Half day 6 h
nothing matched		Full day 12 h	5	Hours 10–12 <i>and</i> orders ≥ 6	Actual hours
			6	Hours 10–12 <i>and</i> orders < 6	Half day 6 h
			7	Hours > 12	Full day 12 h
			nothing matched		Actual hours

Both lists are editable on screen. **Restore SOP starting rules** puts them back.

6. Worked days — what the grid must show

Client	Zone band	Orders	Logged hours	Decided by	Day · credited
Americana	Any	0	–	Rule 1	Absent · 0 h
Americana	Any (Khiran)	3	9.0	Rule 2	Full day · 12 h
Americana	Low	4	8.0	Rule 3	Reduced day · 3 h
Americana	Low	5	8.0	Default	Full day · 12 h
Americana	Good	6	8.0	Rule 4	Reduced day · 3 h
Americana	Good	7	8.0	Default	Full day · 12 h
Keeta	—	0	0.0	Rule 1	Absent · 0 h
Keeta	—	2	9.0	Rule 2	Absent LO · 0 h
Keeta	—	8	3.0	Rule 3	Absent LH · 0 h
Keeta	—	6	5.0	Rule 4	Half day · 6 h
Keeta	—	8	11.0	Rule 5	Actual hours · 11 h
Keeta	—	4	11.0	Rule 6	Half day · 6 h
Keeta	—	8	13.0	Rule 7	Full day · 12 h
Keeta	—	6	8.0	Default	Actual hours · 8 h

Required Hours and Attendance Efficiency

Required Hours = (days in the month – Off Structure) × the full-day hours. **Attendance Efficiency** = Actual Worked Hours ÷ Required Hours, uncapped.

Month length	Off Structure	Required hours	Actual hours	Efficiency
31 days	2	348	300	86.2%
31 days	4	324	300	92.6%
30 days	2	336	336	100.0%
30 days	2	336	400	119.0%

Off Structure changes **Required Hours only**. It never changes the day cells — an OFF cell still comes from approved leave.

7. How to verify — Admin

Sign in at <https://dpdadmin-prod.vercel.app> and open **Payroll & Requests** (/en/payroll). You need a role holding `payroll.view`; edit rights need `payroll.manage` and the export needs `payroll.export`.

A. The month and the filters

- Five tabs are present: **Payroll** · **Combined Payroll** · **Attendance & Orders** · **Requests** · **Settings**.
- Range pills: **This month** · **Last month** · **Custom**. Custom accepts a real From / To range, but only inside this month and the previous two.
- The **Partner view** row lists each client with its rider count. Selecting one narrows the page — and the client is the same thing as the rider's Partner.
- **Zone category** chips (Good / Average / Low / Not set) filter the riders the same way.
- The summary strip reads *Partner · riders · Good / Avg / Low / Not set · rules use zone + orders*.
- Clicking a legend chip (Full day, Absent, 3 h ...) filters the page to riders who have that day; the share card appears; clicking again clears it.

B. The Payroll tab

- KPI row: **Riders in payroll** · **Total active riders** · **Avg attendance eff.** · **Riders at / above 100%** · **Riders with unjustified days** · **3 h Days** · **Manual Adjustments**.
- Per-rider columns include Off Structure, Required Hours, Actual Hours, Efficiency%, Zone band, and a count per day type.
- Column filters work per column (text, a list with an explicit All, and a min/max range) and **Clear column filters** resets them.
- **Export to Excel** downloads the filtered rows exactly as shown.

C. Combined Payroll — the spreadsheet grid

- Hovering the grid shows the hint: *drag across cells to select · drag the corner handle to fill · Ctrl+C, Ctrl+V, Ctrl+D*.
- Drag across a block of a rider's days. The selection bar reads *N cells selected · R riders × D days*.
- Drag the **green corner handle** downwards to fill the pattern into the cells you drag over. The target cells show a dashed preview before you release.
- Select a block and press **Ctrl+C**, then select another block and **Ctrl+V**. A smaller copied block tiles across the selected block. **Ctrl+D** fills down. **Esc** clears the selection.
- Select several cells and press **Apply**. The adjustment dialog opens with the cell / rider / day counts, the new status, and a **Reason** box.
- **Apply with an empty reason is refused**. The reason is mandatory for every manual adjustment.
- Choose **Keep selected status** to leave the cells exactly as the rules computed them, or pick a status (Full day 12 h, Reduced 3 h, Half day 6 h, Actual, OFF, Absent LH / LO, Sick, Accident, Vehicle issue, Custom hours).
- After applying, the changed cells carry an **orange marker**, and the **Manual changes record** shows the date/time, the user, the rider, the day, the original value, the adjusted value and the reason.
- Press **Recompute** / reload the month. The hand-adjusted cells must **not** change — a later recalculation never overwrites a manual adjustment.

D. Off Structure

- The header button **Off Structure** opens the dialog titled *Contracted OFF days for <month>*.
- Pick one rider, set the days and **Save**. The line under the picker shows the current value and its source (default / manual / bulk upload).
- **Download template**, fill a sheet with a Driver ID (or Employee Number) column and an **Off days** column, then **Upload employee sheet**.
- The preview lists every row with its verdict: **Will apply** · **No change** · **Missing ID** · **Duplicate** · **Invalid off days** · **Exceeds month** · **Unknown ID** · **Ambiguous ID** · **Wrong month**.
- **Apply valid rows** writes only the real changes. A sheet that changes nothing answers *No Off Structure changes to write. Default OFF = 2 stays implicit*.
- Set a rider to 4 OFF and re-check the Payroll tab: **Required Hours** drops by $4 \times 12 = 48$ h and Efficiency rises. The **day cells do not move**.

E. Settings — the rules engine

- Open **Settings**. The two seeded clients appear with their criteria ticks and their rule list for the month.
- The rule table reads as sentences — *Zone band is one of Good, Average · Orders is less than 7 → Reduced day · 3 h* — with the credited hours beside each result.
- **Add rule**, add conditions, reorder with the up/down arrows, and **Save rules**. The list is evaluated top to bottom.

- ☐ **Try the rules** is not saved anywhere. Enter a zone, a band, orders and logged hours, and it answers which rule decided the day and what it credited — or that no rule matched and the client default applies.
- ☐ Change the client criteria (hour values, Required hours / day, Off / 30, Good zone $\geq$ %, Average zone $\geq$ %, and the default status) and **Save client**.
- ☐ **Good must be at least Average**. A Good of 60 with Average of 70 is refused.
- ☐ Hours outside 0–24 are refused.
- ☐ **Add client** creates a new client, optionally copying another client's starting rules. A duplicate name is refused.
- ☐ **Delete client** refuses when riders are still assigned to it, and refuses outright for a system client.
- ☐ **Restore SOP starting rules** puts the seeded SOP list back.

F. Settings — Zones & efficiency

- ☐ The banner repeats that the figures belong to the **previous completed month**, not the selected one.
- ☐ Each zone shows Orders · Rider-days · DPD · Target DPD · Efficiency (auto) · Automatic category.
- ☐ **Recompute from last month** refreshes the whole table from the previous completed month.
- ☐ Override the **DPD used**, the **target used** or the **final category** for one zone and press **Use**. The row gains an **Edited** tag and the **Final category** is what the rules read. The automatic figure stays visible beside it.
- ☐ Re-run **Recompute**. Your override must survive.
- ☐ The **(no zone)** row is synthetic — it can be read but not saved.
- ☐ Change **Good $\geq$ % / Average $\geq$ %** and confirm the automatic banding of every zone moves accordingly.

G. Change log

- ☐ Every action above appears in the **Change log** with When · By · Client · Action · Detail — client created, updated, recomputed, adjusted, reset, month opened.
- ☐ Make one change and confirm the row carries your own name and the current time.

H. Permissions

- ☐ A user with `payroll.view` only: no **Off Structure** button, no adjustment dialog, no save button in Settings, no **Export**.
- ☐ A user with `payroll.manage` but not `payroll.export` : can edit, cannot export.
- ☐ A user with neither: the module is not reachable.

8. How to verify — Rider app (User side)

Payroll is an admin screen only. There is no rider-facing payroll page in this release and no rider app update to publish. The rider side is the *input* to payroll: what a rider clocks, delivers and asks for. Verify those three inputs land on the admin grid.

- ☐ **Clock-in / clock-out.** With a test rider, clock in at a known time and clock out a few hours later. On **Admin** → **Payroll**, that rider's **Actual Hours** for that date must equal the check-in → check-out hours.
- ☐ **Orders.** Have the rider submit an order and then verify it on Admin → Deliveries. The verified order must appear as that rider's **Final orders** for that Kuwait day, and it is the number the client rules read.
- ☐ **Requests.** Raise a **Leave** (or Sick Leave / Accident) request from the app. It shows as Pending in the app. On Admin → Payroll that day must still be *unjustified*.
- ☐ Approve the request on Admin → Requests. Come back to Payroll: the day is now **OFF** (or Sick / Accident), it credits 0 hours, and the rider is no longer counted in **Riders with unjustified days**.
- ☐ **Precedence, seen from the rider side.** Approve the leave, then hand-adjust that same day in the Combined grid. The hand adjustment wins, and the cell keeps the orange marker.
- ☐ **No surprise movement.** A client whose rule list has not been saved for the month keeps the previous behaviour — a clocked-in day is a full 12 h day. Confirm on a client you have not configured yet.

9. Notes and limits worth stating

- **No rider app release is required for any of this.** The module is admin-only.
- The month window is **this month plus the previous two**. That is the existing payroll bound and is unchanged in v4; widening it is a separate decision.
- Zone efficiency is **read from the previous completed month** by design, so a newly created zone is **Not set** until a full month has passed and you press Recompute.
- One adjustment batch can cover at most **2,000 cells**, and one Off Structure upload at most **2,000 rows**. Split anything larger.
- A hand adjustment without a reason cannot be saved — that is the rule that makes the change log auditable.
- Nothing in this release recalculates a month you have already paid. Past days keep the value they were given; only the rules and bands you apply from now on decide new days.